

LAND SUBSIDENCE SUSCEPTIBILITY MAPPING AND SPATIO-TEMPORAL FORECASTING BY INTEGRATING SENTINEL-1 INSAR IMAGERY WITH MACHINE AND DEEP LEARNING MODELS

A Thesis

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This thesis is dedicated to me

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The is my Acknowledgments

Table of Contents

Acknowledgments	iii
List of Tables	v
List of Figures	vi
Abstract	vii
Chapter 1. Introduction	1
1.1. Background	1
1.2. Thesis Objectives	3
1.3. Thesis Organization	3
Chapter 2. Study Area and Materials	5
2.1. Study Area	5
2.2. Materials	8
Chapter 3. Literature Review	11
3.1. Overview of Land Subsidence in Urban Areas	11
3.2. Advancements in InSAR Monitoring for Subsidence	12
3.3. Machine Learning for Subsidence Susceptibility Mapping	14
3.4. Deep Learning and Time Series Forecasting of Subsidence	16
3.5. Subsidence Driving Factor Analysis and Explainability	17
3.6. Data Scarcity and Augmentation Techniques	18
3.7. Research Gaps and Study Objectives	18
Chapter 4. Methods	20
4.1. Methodologies	20
Chapter 5. References	27

List of Tables

List of Figures

2.1	Location map of East Baton Rouge Parish	7
4.1	Overall methodology flowchart	21
4.2	Random Forest model architecture	24
4.3	Extra Trees model architecture	25

Abstract

Land subsidence has become an escalating geohazard in rapidly urbanizing regions such as East Baton Rouge Parish (EBRP), Louisiana, where groundwater extraction, infrastructure loading, and geomorphological variability jointly drive ground deformation. This study presents an integrated framework for subsidence susceptibility mapping and spatio-temporal forecasting by combining Sentinel-1 SBAS-InSAR observations with machine learning and deep learning models. Multi-temporal SBAS-InSAR data spanning 2017-2025 (246 descending scenes) were analyzed alongside fifteen geospatial conditioning factors (geology, LULC, DEM, slope, aspect, TPI, curvature, TWI, precipitation, and distances to rivers, roads, faults, railways, bridges, and groundwater wells). Gaussian noise-based data augmentation was applied to enhance model generalization. Susceptibility mapping identified Extra Trees Regressor as the optimal spatial model, achieving test R^2 of 0.9063 versus Random Forest ($R^2 = 0.8037$). A physics-informed LSTM approach with adaptive stabilization forecasted subsidence with R^2 of 0.9368 and Pearson correlation of 0.9723. SHAP analysis highlighted LULC, fault proximity, and River-DEM interactions as dominant drivers. Projections to 2030 indicate gradual stabilization, with mean velocity of -0.530 mm/year and cumulative displacement of -6.42 mm.

Chapter 1. Introduction

1.1. Background

Land subsidence is a gradual or sudden downward displacement of the Earth’s surface caused by both natural and anthropogenic processes. While natural mechanisms such as sediment consolidation, glacial isostatic adjustment, and tectonic motion contribute to subsidence, human activities are now recognized as the dominant drivers in most severe global cases, accounting for more than 80% of documented high-impact events worldwide (Higgins, 2016). . Excessive groundwater extraction, aquifer-system compaction, soil consolidation, and static loading from urban infrastructure significantly increase subsidence rates, posing substantial risks to infrastructure stability, groundwater sustainability, flood resilience, and environmental systems in rapidly urbanizing regions. East Baton Rouge Parish (EBRP), Louisiana, represents a prominent subsidence-prone urban area where geological, hydrological, and anthropogenic factors interact in a complex manner. The region is characterized by low-relief topography and unconsolidated sediments, making it highly sensitive to pore-pressure reduction caused by intensive pumping from the Southern Hills Aquifer System. Rapid urban expansion and heavy infrastructure loading further exacerbate ground deformation. Structural controls such as the Baton Rouge and Denham Springs fault systems introduce spatially heterogeneous and non-linear deformation patterns, increasing vulnerability to flooding, pipeline rupture, and foundation failure (Abdalla et al., 2024; L. Wang et al., 2021).

Traditional land-subsidence monitoring techniques, including geotechnical surveys, levelling campaigns, and GNSS/GPS measurements, provide high accuracy at discrete locations but are limited by sparse spatial coverage, high operational costs, and insufficient

temporal resolution for regional-scale risk assessment. The advent of satellite-based Interferometric Synthetic Aperture Radar (InSAR) has transformed subsidence monitoring by enabling millimetre-level deformation measurements over large areas. In particular, time-series approaches such as Persistent Scatterer InSAR (PS-InSAR) and Small Baseline Subset InSAR (SBAS-InSAR) have proven effective for detecting subtle, long-term deformation in urban and mixed land-cover environments (Berardino et al., 2002; Ferretti et al., 2001, 2011).

Despite these advances, converting large-scale InSAR observations into reliable and interpretable spatio-temporal forecasts remains challenging due to the non-linear, multi-factor nature of subsidence processes. Physically based numerical models offer insight into deformation mechanisms but often require detailed soil and hydrogeological parameters that are rarely available at regional scales (Zhou et al., 2022). Recent developments in machine learning have demonstrated strong potential for addressing these challenges by learning complex relationships between deformation and environmental controls directly from data. In particular, tree-based ensemble models such as the Extra Trees Regressor have shown strong performance in spatial susceptibility mapping due to their robustness to multicollinearity, ability to model non-linear interactions, and reduced overfitting compared to conventional Random Forest approaches (E. Hosseinzadeh et al., 2024).

While ensemble tree models are well suited for spatial susceptibility analysis, they do not explicitly capture temporal dependencies. Long Short-Term Memory (LSTM) networks address this limitation by modelling sequential deformation dynamics and long-term temporal dependencies within InSAR time series. However, purely data-driven LSTM models may generate physically unrealistic trends when extrapolated beyond the observation period. To overcome this limitation, recent studies have emphasized the integration of deep learning

with physics-based constraints to improve forecast stability and geomechanical plausibility (Zhu et al., 2024).

Guided by this framework, the present study integrates SBAS-InSAR-derived deformation, Extra Trees-based spatial susceptibility mapping, and a physics-informed LSTM spatio-temporal forecasting approach enhanced with explainable artificial intelligence. This integrated methodology provides a robust, interpretable, and geomechanically consistent framework for assessing present-day subsidence patterns and generating reliable projections of ground deformation in East Baton Rouge Parish through 2030.

1.2. Thesis Objectives

The main research objectives are to:

- i. Generate high-resolution land-subsidence map for East Baton Rouge Parish using Sentinel-1 SBAS-InSAR data (2017–2025)
- ii. Identify and rank key environmental and anthropogenic drivers for land subsidence using EXtra Trees and Random Forest machine-learning model with SHAP-based Explainability.
- iii. Develop a physics-informed LSTM-based spatio-temporal model to forecast land subsidence from InSAR time-series data, ensuring geomechanically consistent projections through 2030.

1.3. Thesis Organization

This thesis is structured into six chapters as follows:

Chapter 1, this chapter, introduces the research topic of land subsidence susceptibility mapping and spatio-temporal forecasting. It provides background information, outlines research objectives, and presents the thesis organization.

Chapter 2 presents the study area characteristics of East Baton Rouge Parish, including its geological setting, hydrogeological conditions, and historical subsidence patterns that

make it vulnerable to ground deformation.

Chapter 3 provides a comprehensive literature review covering: advances in InSAR monitoring techniques for subsidence, machine learning applications in susceptibility mapping, deep learning approaches for time-series forecasting, hybrid modeling frameworks, and methods for analyzing subsidence driving factors. The chapter concludes by identifying research gaps this study aims to address.

Chapter 4 details the methodology employed, including: SBAS-InSAR processing of Sentinel-1 data, preparation of 15 environmental and anthropogenic predictor variables, development of machine learning models (Random Forest and Extra Trees) for susceptibility mapping, implementation of physics-informed LSTM for temporal forecasting, and SHAP-based interpretation methods for explainability.

Chapter 5 presents the analysis results and discussion, covering: spatial patterns of observed subsidence, relative importance of driving factors, performance evaluation of individual and ensemble models, and validation against ground-truth measurements. The implications for urban planning and groundwater management are also discussed.

Chapter 6 summarizes key conclusions, provides recommendations for land subsidence monitoring and mitigation in East Baton Rouge, acknowledges study limitations, and suggests directions for future research.

Chapter 2. Study Area and Materials

2.1. Study Area

This study is conducted in East Baton Rouge Parish (EBRP), located in southeastern Louisiana, United States, within the Baton Rouge Capital Area. The parish covers approximately 471 square miles and extends between 30°18'–30°36'N latitude and 91°06'–90°48'W longitude (Figure 2.1). The study area is bounded by major hydrological systems, including the Mississippi River to the west and the Comite River to the east. It is characterized by low-relief floodplains, forested wetlands, and extensive alluvial deposits typical of the Gulf Coast region. This geographic setting places East Baton Rouge Parish within a broader zone of documented subsidence and aquifer-system response associated with sedimentary basins in the southern United States (Huning et al., 2023).

Geologically, East Baton Rouge Parish is underlain by thick sequences of unconsolidated Holocene and Pleistocene deltaic and fluvial sediments, composed primarily of clay, silt, and fine sand. These young, poorly consolidated materials exhibit high compressibility and low shear strength, making the region particularly susceptible to land subsidence and soil deformation when subjected to changes in effective stress (Higgins, 2016). Such geological conditions are characteristic of sedimentary basin and coastal plain environments, where compaction-driven subsidence is commonly observed.

Land subsidence in EBRP is driven predominantly by anthropogenic activities, especially long-term groundwater extraction from the Baton Rouge aquifer system. Sustained withdrawal of groundwater reduces pore pressure within aquifer systems, leading to aquifer-system compaction and progressive lowering of the land surface (Huning et al., 2023). These

effects may be irreversible where inelastic compaction occurs. Subsidence associated with groundwater depletion has been widely documented across sedimentary basins in the United States and is increasingly mapped using geodetic techniques (Higgins, 2016).

In addition to groundwater withdrawal, urban development and surface loading contribute to subsidence processes within East Baton Rouge Parish. Expansion of infrastructure and increased building density impose additional loads on compressible sediments, accelerating consolidation in low-lying areas. The combined effects of fluid withdrawal and surface loading produce complex deformation patterns that vary spatially and temporally, emphasizing the need for spatially continuous monitoring approaches such as Interferometric Synthetic Aperture Radar (InSAR) (Crosetto et al., 2016; Ferretti et al., 2001).

Climatic and hydrological conditions further influence ground deformation in the study area. East Baton Rouge Parish experiences a humid subtropical climate, with high annual precipitation, intense rainfall events, and periodic flooding. Seasonal variations in groundwater levels, soil moisture, and surface water storage can induce transient and non-linear deformation signals, which may be superimposed on long-term subsidence trends. These effects complicate the interpretation of InSAR time series and require careful consideration of atmospheric and hydrologic contributions to observed displacement (Berardino et al., 2002; Fattahi & Amelung, 2015).

Given the combined influence of compressible deltaic sediments, intensive groundwater extraction, urban loading, and hydro-climatic variability, East Baton Rouge Parish represents an ideal natural laboratory for investigating land subsidence using Sentinel-1 time-series InSAR. The spatial extent of the study area supports high-resolution deformation mapping

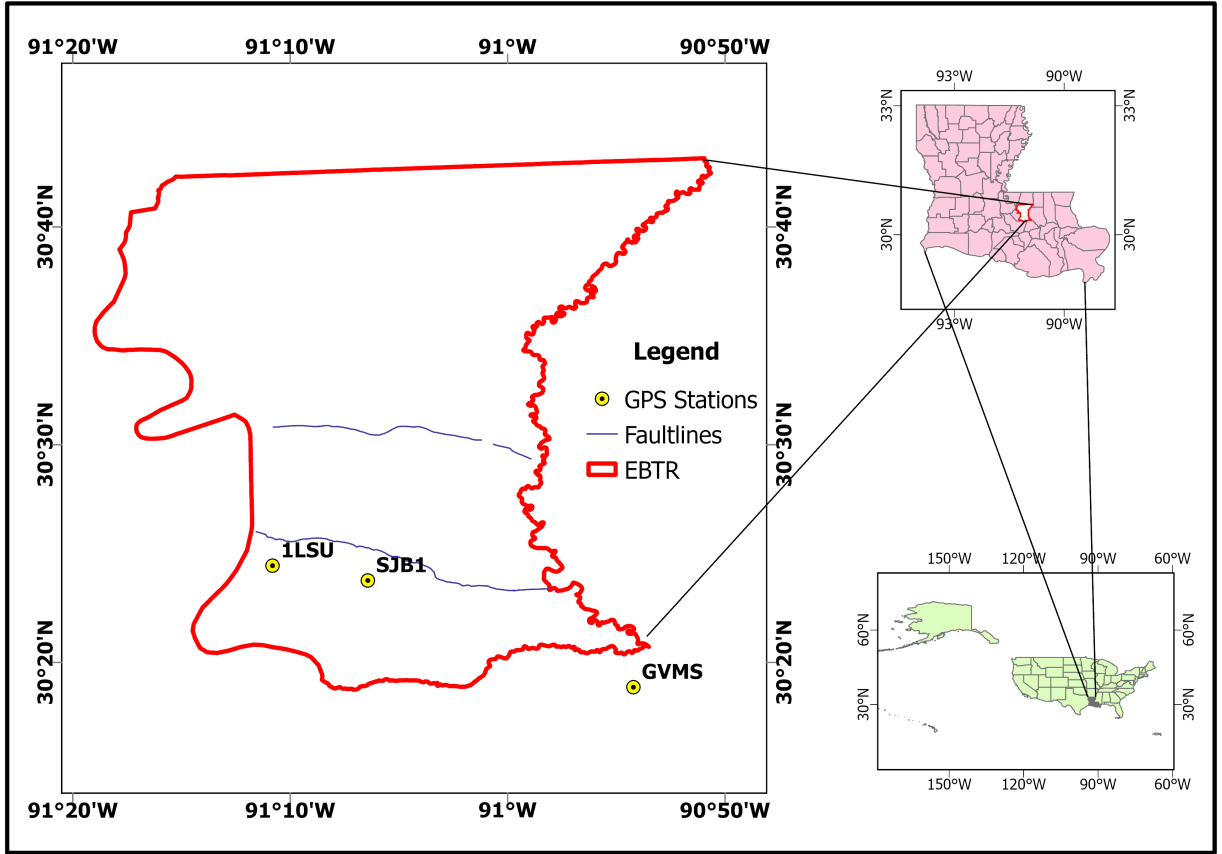


Figure 2.1. Location map of East Baton Rouge Parish

and facilitates integration with GNSS control points for validation and uncertainty assessment. Improved understanding of the spatial and temporal characteristics of subsidence in EBRP is essential for sustainable groundwater management, infrastructure resilience, and long-term planning in sedimentary basin environments (Crosetto et al., 2016; Higgins, 2016).

Having established the geophysical and environmental context of East Baton Rouge Parish, the following chapter reviews the current state of knowledge in InSAR-based subsidence monitoring, machine learning susceptibility mapping, and temporal forecasting approaches that form the methodological foundation of this study.

2.2. Materials

This study utilises a comprehensive set of satellite data, auxiliary geospatial datasets, atmospheric products, and open-source software tools to drive ten (8) years (2017-2025) of ground deformation and velocity estimates using timeseries SBAS-InSAR techniques. The materials used are described in detail below.

2.2.1. Sentinel-1 SAR Dataset

A total of 246 Sentinel-1A descending line-of-sight (LOS) Single-Look Complex (SLC) scenes were used in this study. The data were acquired in Interferometric Wide (IW) mode with VV polarisation in descending tracks. Sentinel-1 is a C-band (5.405 GHz) SAR mission operated by the European Space Agency (ESA) and provides systematic, long-term observations suitable for deformation monitoring.

2.2.2. Precise Orbit Ephemerides

Precise Orbits Ephemerides (POEORB) provided by the ESA were applied to all Sentinel-1 acquisitions prior to interferometric processing. Compared to restituted orbits, POEORB significantly improves satellite position accuracy, reducing residual orbital errors and mitigating long-wavelength phase ramps in interferograms (Ferretti et al., 2001).

2.2.3. Digital Elevation Model and Atmospheric Data

A 30 m resolution Digital Elevation Model (DEM) was used to correct for topographic phase contributions in interferograms generation. The DEM also served as the basis for deriving topographic conditioning factors used in the susceptibility analysis.

To mitigate atmospheric artefacts, particularly stratified tropospheric delays, ERA5 (Fifth Generation ECMWF Atmospheric Reanalysis) data were incorporated during timeseries

processing. ERA5 provides high-resolution global estimates of atmospheric state variables and has been widely adopted for atmospheric phase screen correction in InSAR studies (S. R. Hosseinzadeh et al., 2024).

2.2.4. Geospatial Conditioning Factors

A total of fifteen (15) geospatial conditioning factors were prepared at 30 m resolution and aligned to a common spatial grid for land subsidence susceptibility modelling. These factors were grouped into topographic, geological, hydrological, structural/tectonic, infrastructure, climatic, and anthropogenic categories, reflecting the multi-causal nature of land subsidence processes.

- Topographic factors: Digital Elevation Model (DEM), slope, aspect, Topographic Position Index (TPI), curvature, Topographic Wetness Index (TWI).
- Geological factors: geology, Land Use/Land Cover (LULC).
- Hydrological factors: distance from river.
- Structural/Tectonic factors: distance from fault (Baton Rouge and Denham Springs fault systems).
- Infrastructure factors: distance from road, distance from railway, distance from bridge.
- Climatic factors: precipitation.
- Anthropogenic factors: well_idw (interpolated groundwater well influence).

Continuous layers were min–max scaled; categorical layers (geology, LULC) were one-hot encoded.

2.2.5. Validation Data

InSAR-derived deformation was validated using GNSS vertical displacement time series from the GulfNet CORS stations (1LSU and GMVS) in EBRP obtained from the Nevada

Geodetic Laboratory (NGL). GNSS provides high-precision point-based measurements and serves as a standard benchmark for InSAR timeseries validation (Abdalla et al., 2024).

With the study area characteristics and data sources established, Chapter 3 examines the existing literature on InSAR monitoring techniques, machine learning applications in subsidence analysis, and identifies the research gaps that motivate this integrated methodological approach.

Chapter 3. Literature Review

3.1. Overview of Land Subsidence in Urban Areas

Land subsidence is increasingly recognized as a major global geohazard, with over 80% of severe cases associated with aquifer compaction and groundwater withdrawal (Higgins, 2016). In rapidly urbanizing regions like the Lower Mississippi Basin, the interplay between shallow geology and heavy infrastructure growth intensifies surface deformation. Urban environments are particularly susceptible due to the high density of man-made structures that act as static loads on compressible soil layers. Research indicates that unmitigated subsidence triggers a cascade of secondary hazards, including pipeline ruptures, building tilting, and structural failures, resulting in significant economic losses (Zou et al., 2016).

Land subsidence, which is defined as the slow downward movement of the Earth's surface, is becoming more widely acknowledged as a significant geohazard in areas of the world that are rapidly becoming more urbanized (Huning et al., 2023). In urban environments, subsidence processes are fueled by excessive groundwater extraction, sediment consolidation, urban infrastructure loading, and land-use change (Holzer & Johnson, 1985). Research on the Lower Mississippi Basin, for instance, demonstrates how shallow geology and infrastructure growth can intensify surface deformation and compaction (Zou et al., 2016). The hazards include diminished drainage capacity, heightened susceptibility to flooding in low-lying urban corridors, and harm to pipelines and foundations. Subsurface faulting and structural geology make subsidence patterns in many cities even more complex (Younas et al., 2023). These findings highlight the need for reliable frameworks that can both map the areas where subsidence is most likely to occur and predict its future course.

3.2. Advancements in InSAR Monitoring for Subsidence

The development of Synthetic Aperture Radar Interferometry (InSAR) has transformed ground-deformation monitoring by providing high-resolution, wide-area observations. Traditional geodetic techniques, such as GNSS and levelling, have historically provided precise point measurements but are limited by prohibitive costs and low spatial resolution.

3.2.1. Evolution of InSAR Techniques for Subsidence Monitoring

InSAR emerged in early 1990s as a transformative remote sensing technique capable of detecting ground deformation by exploiting phase differences between radar images acquired at different times. Because radar phase is sensitive to path-length changes on the order of millimetres, InSAR enables precise measurement of surface displacement over large spatial extents (Jiang et al., 2023).

Early applications relied on Differential InSAR (D-InSAR), which isolates deformation by subtracting topographic phase contributions using DEM. D-InSAR proved effective for mapping rapid and spatially coherent deformation associated with earthquakes and volcanic activity (G. Wang et al., 2024). However, its application to long-term subsidence monitoring is limited by atmospheric phase delay, temporal decorrelation, orbital errors, and phase unwrapping ambiguities, particularly over vegetated or urbanizing surfaces (Hussain et al., 2023).

To overcome these limitations, Persistent Scatterer InSAR (PS-InSAR) was introduced to exploit radar targets that remain phase-stable over long time periods, such as buildings and exposed infrastructure (Ferretti 2001). By analyzing large stacks of SAR acquisitions,

PS-InSAR achieves sub-millimeter precision on built infrastructure (Ferretti et al., 2001, 2011). However, PS-InSAR is often sparse in non-urbanized or vegetated areas, limiting its applicability to mixed land-cover environments.

Small Baseline Subset (SBAS-InSAR) has emerged as a superior alternative, particularly in regions mixed urban and vegetated terrain (Berardino et al., 2002). It was developed to improve spatial coverage by forming interferogram networks with short temporal timeseries reconstruction over distributed scatterers, making it particularly suitable for regions characterized by mixed land cover. By utilizing multiple master interferograms with small perpendicular baselines, SBAS identifies Slowly Decorrelating Filtered Pixels (SDFP), which significantly reduces the impact of spatial and temporal decorrelation. Recent studies indicate that large-scale time-series data processing supports the identification of both linear and complex non-linear deformation behaviours with millimetre-level accuracy. Comparatively, SBAS-InSAR provides up to 2.83 times higher monitoring density than PS-InSAR in mixed urban-rural environments (Bayik et al., 2025).

Although SBAS-InSAR generally exhibits lower point-wise precision than PS-InSAR, it provides dense and spatially continuous deformation fields essential for regional subsidence analysis (Crosetto et al., 2016).

Unlike traditional geodetic methods, InSAR enables millimeter-scale motion detection across vast regions regardless of weather or lighting conditions (Crosetto et al., 2016). Building on these advances, newer methods such as Distributed Scatterer (DS-InSAR) and Multi-Temporal InSAR (MT-InSAR) extend coherence to mixed urban-rural environments by statistically analyzing pixel clusters (Fattahi & Amelung, 2015).

More recent approaches, including Distributed Scatterer (DS-InSAR) and hybrid

methods such as SqueeSAR, further extend coherence in low-reflectivity areas by statistically analysing homogeneous pixel clusters (Ferretti et al., 2011)

3.2.2. Temporal Decorrelation, Atmospheric Effects, and Time-Series Reliability

Temporal decorrelation remains a fundamental challenge across all InSAR techniques, arising from changes in surface scattering properties caused by vegetation growth, soil moisture variation, and anthropogenic activity (Zheng et al., 2024). In urban–natural transition zones, heterogeneous coherence leads to spatially variable uncertainty in deformation estimates.

Atmospheric phase delay, particularly tropospheric water vapour variability, introduces long-wavelength artefacts that can mimic real deformation signals. These effects are especially pronounced in humid subtropical regions such as the U.S. Gulf Coast. Recent integration of numerical weather models and reanalysis products, including ERA5 and GACOS, has significantly improved atmospheric correction and time-series reliability

3.3. Machine Learning for Subsidence Susceptibility Mapping

Machine learning (ML) has become the global standard for modelling the non-linear relationship between ground deformation and various geospatial drivers, such as Land Use/Land Cover (LULC), soil types, and distance from faults. Gharechae et al. (2023) applied RF, KNN, and CART to InSAR-derived subsidence in Iran’s Bakhtegan Basin, finding RF performed best and highlighting that the leading susceptibility factors were not only groundwater withdrawal but also proximity to dams, faults and mines. Orlandi et al. (2024) compared ML approaches in Murcia, Spain, using land-use, lithology, and other predictors, demonstrating the utility of ML for zonation and factor ranking.

3.3.1. Ensemble Learners and Extra Trees Regression

While Random Forest (RF) remains a common benchmark due to its ability to handle multi-variate data without pre-defined assumptions, it can be prone to overfitting in complex geological datasets (Breiman, 2001). This study identifies the Extra Trees (ERT) Regressor as a more robust alternative for subsidence susceptibility mapping. By implementing random splits at the node level, ERT enhances model generalisation and reduces variance compared to conventional bagged ensembles (Yang et al., 2022). In the East Baton Rouge Parish region, Extra Trees achieved a test R^2 of 0.9063, significantly outperforming Random Forest ($R^2 = 0.8037$), demonstrating superior predictive power and reduced overfitting for complex, multi-factor subsidence systems.

3.3.2. Gaussian Process Regression and Data Augmentation

Gaussian Process Regression (GPR) is increasingly used for its ability to model calibrated predictive uncertainty, making it highly effective for small datasets and areas with limited observability. To address the technical challenge of small dataset sizes in local subsidence surveys, Gaussian noise-based data augmentation has been validated as an effective strategy to expand training samples and ensure model stability (Yang et al., 2022). This augmentation approach mitigated data gaps and enhanced model generalization, expanding the effective training dataset to over 390,000 samples. S. R. Hosseinzadeh et al. (2024) evaluated seven ML methods (CART, BRT, SVM, RF, logistic regression, etc.) in Iran and found BRT outperformed others in susceptibility mapping. These studies demonstrate ML's strength in spatial patterning and factor ranking, yet many stop short of integrating temporal forecasting or sequence modelling, limiting their value for prediction of future

subsidence trends.

3.4. Deep Learning and Time Series Forecasting of Subsidence

Deep-learning (DL) models, especially sequence models like LSTM, are increasingly adopted for subsidence forecasting because of their ability to capture temporal dependencies and non-linear dynamics. Li et al. (2020) applied LSTM to PS-InSAR data in Beijing Plain, showing promise for temporal modelling though with limitations in spatial generalization. Mirmazloumi et al. (2023) used LSTM on InSAR time-series for three mining sites, demonstrating an early-warning capability by forecasting 3–5 steps ahead.

3.4.1. Physics-Informed Neural Networks (PINNs)

Purely data-driven models may produce physically implausible results, such as unrealistic acceleration or velocity reversals. Physics-Informed Machine Learning (PIML) frameworks integrate geophysical constraints (e.g., velocity continuity and conservation laws) to ensure that long-term projections remain geomechanically consistent. Raissi et al. (2019) developed Physics-Informed Neural Networks (PINNs), which encode physical laws as penalty terms in the loss function, thereby constraining the model to produce physically realistic predictions. Such gaps suggest the potential for stacked models using meta-learning ensembles to combine base-model predictions and latent features for enhanced accuracy and interpretability. Hybrid models are common in other geohazard domains (e.g., landslides, groundwater) but remain under-utilized in urban subsidence. Thus, implementing a hybrid DL/ML ensemble to both map and forecast subsidence fills a methodological gap in current literature.

3.5. Subsidence Driving Factor Analysis and Explainability

A critical limitation of traditional deep learning is its “black-box” nature, which hinders its use in urban planning and policy-making. Explainable AI (XAI) tools, specifically SHapley Additive exPlanations (SHAP), are now applied to quantify the marginal contribution of each input factor to the subsidence signal. Yu et al. (2025) employed SHAP to interpret XGBoost-LightGBM-CatBoost ensemble models for land subsidence susceptibility mapping in Hangzhou, demonstrating that engineering activities and soft soil distribution are the most influential factors. Xue et al. (2024) used ML on InSAR data to inspect deformation along high-speed rail corridors, emphasizing hydrological and structural controls.

Understanding why subsidence occurs and identifying and quantifying drivers is vital for decision-making in resource management and infrastructure planning. SHAP analysis has revealed that while LULC and fault proximity are primary contributors, River-DEM interactions and groundwater fluctuations account for significant portions of the deformation signal in regional contexts. By integrating SHAP within a hybrid forecasting pipeline, transparent and scientifically defensible explanations of subsidence mechanisms can be provided, supporting sustainable water resource and infrastructure management (Yu et al., 2025). These studies show how driver-ranking can support mitigation strategies and inform policy decisions. Nevertheless, few investigations combine hybrid DL forecasting with explicit causative-factor analysis in urban contexts; the use of SHAP in a hybrid DL/ML framework addresses a real literature gap.

3.6. Data Scarcity and Augmentation Techniques

In geohazard research, obtaining large datasets for model training is often time-consuming and expensive. Gaussian noise-based data augmentation is a validated strategy to improve model stability and generalisation. Adding small amounts of noise allows the training set to expand significantly without distorting the original data distribution, thereby preventing overfitting in tree-based and deep-learning architectures. This approach has been successfully applied in the context of subsidence prediction with Extra Trees models, where synthetic data generation improved model robustness without compromising prediction accuracy (Yang et al., 2022).

3.7. Research Gaps and Study Objectives

While many studies have applied InSAR and machine or deep learning (ML/DL) methods to analyze land subsidence, most have treated spatial susceptibility mapping and temporal forecasting as separate tasks, often overlooking model interpretability (Gharechaei et al., 2023; Hussain et al., 2023; Rahmani et al., 2024). Applications within complex U.S. Gulf Coast urban environments also remain limited. Recent research by He et al. (2025) highlighted the specific need for integrated monitoring and assessment frameworks in coastal subsidence-prone regions. To address these gaps, this study integrates SBAS-InSAR velocity data with environmental conditioning factors using Extra Trees and Random Forest for spatial subsidence susceptibility mapping, alongside a physics-informed LSTM framework for temporal forecasting. SHAP-based interpretation is employed to develop predictive and explainable subsidence models tailored to Baton Rouge’s environmental and infrastructural characteristics (Abdalla et al., 2024; Mojtahedi et al., 2025).

The identified gaps and methodological advances reviewed in this chapter provide the foundation for the integrated approach detailed in Chapter 4, which combines SBAS-InSAR, ensemble machine learning, and physics-informed temporal forecasting to address subsidence monitoring and prediction in East Baton Rouge Parish.

Chapter 4. Methods

4.1. Methodologies

This chapter details the integrated framework employed to map and forecast land subsidence in East Baton Rouge Parish (Figure 4.1). The methodology combines three complementary approaches: (1) SBAS-InSAR time-series analysis to extract deformation velocity from 246 Sentinel-1 acquisitions, (2) tree-based ensemble learning (Random Forest and Extra Trees) for spatial susceptibility mapping using 15 environmental conditioning factors, and (3) physics-informed LSTM networks for spatio-temporal forecasting. SHAP-based explainability is integrated throughout to identify dominant subsidence drivers and ensure model interpretability.

4.1.1. SBAS-InSAR Processing

SBAS-InSAR was selected over PS-InSAR for this study due to the mixed urban-rural land cover characteristics of East Baton Rouge Parish. While PS-InSAR achieves higher point-wise precision on stable infrastructure, it provides sparse coverage in vegetated and agricultural areas (Ferretti et al., 2011). SBAS-InSAR, by contrast, exploits distributed scatterers and maintains coherence across heterogeneous terrain, yielding 2.83 times higher spatial density in mixed environments (Bayik et al., 2025). This enhanced coverage is critical for regional subsidence mapping and for generating sufficient training samples for machine learning models.

SBAS-InSAR represents a widely adopted time-series InSAR methodology, initially introduced by Berardino et al. (2002). This technique is frequently employed for large-scale urban monitoring applications. The fundamental principle involves constructing a

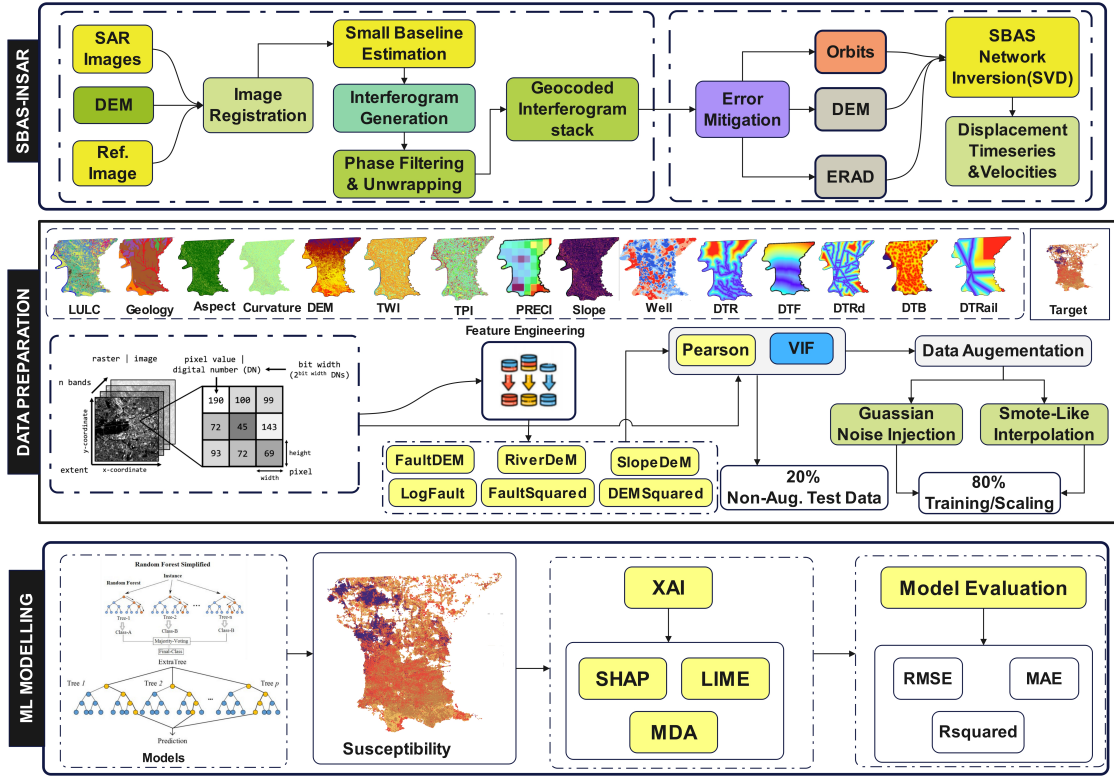


Figure 4.1. Overall methodology flowchart

network of interferometric pairs characterized by minimal temporal and spatial baseline separations derived from the spatial distribution of SAR imagery. This approach as shown in Figure 4.1 facilitates interferogram generation while minimizing decoherence and topographic phase errors. Time-series analysis is subsequently conducted to derive surface deformation information. When M interferograms are generated from $N + 1$ SAR acquisitions, a specific correlation exists between M and N . The constraint governing this relationship is expressed as (Colesanti et al., 2003):

$$\frac{N + 1}{2} \leq M \leq N \frac{N + 1}{2} \quad (4.1)$$

For the i -th differential interferogram constructed within the temporal interval between acquisitions t_A and t_B , the phase corresponding to any pixel within the interferogram can be

formulated as:

$$\begin{aligned}
\delta\phi_i(x, r) &= \phi(t_B, x, r) - \phi(t_A, x, r) \\
&\approx \frac{4\pi}{\lambda}[d(t_B, x, r) - d(t_A, x, r)] + \Delta\phi_{topo}(x, r) \\
&\quad + \Delta\phi_{atm}(x, r) + \Delta\phi_{noise}(x, r)
\end{aligned} \tag{4.2}$$

In this formulation, $\phi(t_A, x, r)$ and $\phi(t_B, x, r)$ denote the phase values at spatial coordinates (x, r) corresponding to times t_A and t_B , respectively. The term $\Delta\phi_{topo}(x, r)$ represents residual topographic phase in the differential interferogram, $\Delta\phi_{atm}(x, r)$ accounts for atmospheric delay phase contributions, and $\Delta\phi_{noise}(x, r)$ represents the decoherence noise phase component. When digital elevation model (DEM) data are available, the topographic phase can be simulated and removed to eliminate terrain-induced phase errors (Fattahi & Amelung, 2015). The phase matrix of all differential interferograms is denoted as $\delta\phi$, while the average surface deformation rate is represented by matrix V . The relationship between these matrices is expressed as:

$$BV = \delta\phi \tag{4.3}$$

The dimensions of matrix B are determined by M and N . In general, matrix B exhibits rank deficiency. Singular Value Decomposition (SVD) is conventionally applied for rank-deficient problem solving. Consequently, the acquisition of surface deformation information in time-series is achieved through the joint solution of multiple baseline sets, employing SVD to obtain the least-squares norm solution (Berardino et al., 2002).

The SBAS-InSAR processing was implemented using the 246 descending-track Sentinel-1 SLC acquisitions spanning 2017–2025. Perpendicular baseline thresholds were set to 150 m and temporal baselines to 48 days to maximize interferometric coherence while ensuring sufficient temporal sampling. Atmospheric phase delays were corrected using GACOS

tropospheric delay models, and phase unwrapping was performed using the Minimum Cost Flow algorithm. The resulting velocity field was geocoded to a 30 m grid aligned with the conditioning factor layers for subsequent machine learning analysis.

4.1.2. Environmental Conditioning Factors

Fifteen geospatial conditioning factors were compiled at 30 m resolution to characterize the environmental and anthropogenic controls on subsidence (as described in Section 2.2.4). These factors span topographic attributes (DEM, slope, aspect, TPI, curvature, TWI), geological characteristics (geology, LULC), hydrological proximity (distance to rivers), structural controls (distance to faults), infrastructure proximity (distances to roads, railways, bridges), climatic variables (precipitation), and anthropogenic activity (interpolated groundwater well influence). Continuous variables were normalized using min-max scaling to $[0,1]$, while categorical variables (geology, LULC) were one-hot encoded. The SBAS-derived velocity values at each 30 m pixel served as the target variable for supervised learning.

4.1.3. Random Forest Model

RF is a supervised machine-learning algorithm that uses several decision trees to produce predictions that are more precise. RF is an ensemble learning technique that produces the mean prediction for regression tasks by combining the predictions of several decision tree models (Figure 4.2) (Wahba et al., 2024; Xi et al., 2023; Zhang et al., 2019). The RF regressor operates an estimator that fits a collection of N randomized decision tree regressors independently and identically distributes random vectors (Breiman, 2001).

The RF training procedure begins by generating random subsets from the original training dataset consisting of independently and identically distributed $[0, 1]^d \times \mathbb{R}$ -valued

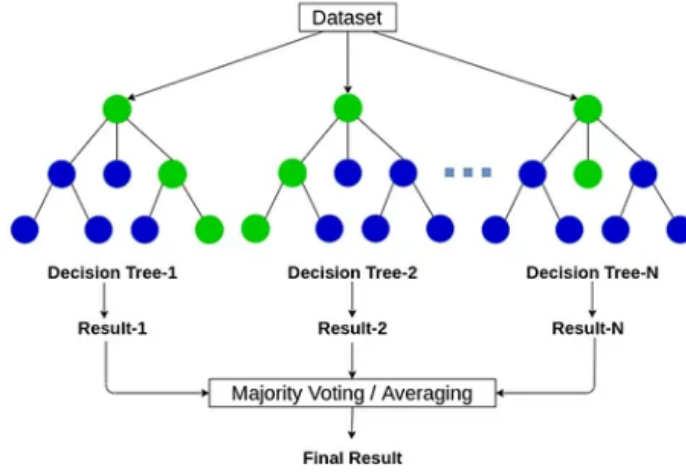


Figure 4.2. Random Forest model architecture

random variables ($d \geq 2$) with comparable distribution properties to the independent generic pairs (x, y) . This is achieved by randomly resampling a specified proportion of the complete dataset $t_n = \{(x_1, y_1), \dots, (x_n, y_n)\}$. This bootstrap aggregating (bagging) technique constructs multiple unpruned decision trees (DTs) (Breiman, 1996; Breiman et al., 2017). Subsequently, K input features are randomly selected for splitting at each node during the training of individual decision trees. The prediction output from the n -th decision tree is represented as $g(x, t_N^n)$, and the aggregate prediction obtained by averaging across all constituent trees forms the final forest predictor, expressed in Equation (5):

$$\hat{y} = \frac{1}{n} \sum_{i=1}^n g(x, t_m^n) \quad (4.4)$$

where $\hat{y}$ represents the final prediction of the Random Forest model.

4.1.4. Extra Trees Algorithm

Both Random Forest and Extra Trees were implemented to identify the optimal ensemble approach for subsidence susceptibility mapping. While RF has been widely applied in geohazard studies, recent research suggests that Extra Trees may achieve superior

generalization in complex, multi-factorial systems (Geurts et al., 2006).

The Extra Trees (ET) algorithm is an ensemble-based method that extends the Random Forest approach. Two fundamental distinctions differentiate ET from RF: (1) ET constructs trees utilizing random subsets of data without replacement, thereby reducing sampling bias, and (2) the randomization in ET originates from random split selection rather than bootstrap aggregation (Geurts et al., 2006). In the ET framework, the tree-splitting mechanism is governed by two hyperparameters: k denotes the number of features randomly selected at each node, and n_{min} specifies the minimum sample size required for node partitioning. These parameters enhance model generalization and mitigate overfitting (Geurts et al., 2006). The architecture of the model is shown in Figure 4.3.

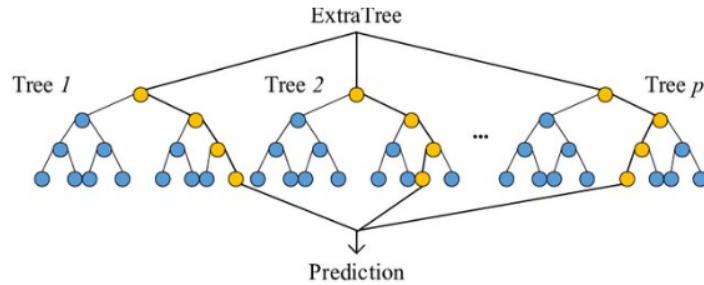


Figure 4.3. Extra Trees model architecture

Analogous to RF, the ET algorithm executes regression through two sequential phases: bootstrapping and bagging. During bootstrapping, a random subset of the training dataset is drawn to construct individual decision trees. In the bagging phase, nodes are partitioned using randomly selected feature subsets. The final prediction is determined by randomly selecting a subset and computing its corresponding output value. Mathematically, the ET

regression process is expressed by the equation provided by Breiman (2001):

$$p(x, \beta_1, \dots, \beta_n) = \frac{1}{m} \sum_{i=1}^m p(x, \beta_n) \quad (4.5)$$

where p represents the p -th prediction tree, and β is a distributed vector.

4.1.5. Gaussian Noise-Based Data Augmentation

To mitigate limited labeled samples, Gaussian noise augmentation was applied by perturbing feature vectors with zero-mean Gaussian noise set to 3% of each parameter’s range. The augmentation expanded the training dataset by a factor of 2.5 to a total of 393,412 samples while preserving overall feature distributions.

Chapter 5. References

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